Time Series Econometrics

Properties of Stationary and Integrated Time Series

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Abstract

This handout reviews properties of stationary and integrated time series, covering definitions, examples, and visual illustrations. It provides a concise reference for students and researchers in time-series econometrics.

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1 Introduction

Time-series econometrics analyzes sequences of observations indexed by time and explicitly models temporal dependence. Time-series data commonly exhibit autocorrelation, non-stationarity, and seasonality. As a result, the standard i.i.d. assumptions often fail and require specialized methods. This handout summarizes core definitions and properties of stationary and integrated processes, and gives concise examples and illustrations.

Distinguishing stationary from integrated series is essential for choosing models and making valid inferences. Central to these ideas is the notion of a stochastic process, defined below.

Definition 1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let T be an index set (often representing time, e.g. $T = \mathbb{N}$ for discrete time or $T = \mathbb{R}_+$ for continuous time).

A stochastic process is a family of random variables

$$\{X(\tau, \omega) : \tau \in T,\ \omega \in \Omega\}$$

such that for each fixed $\tau \in T$, the mapping

$$X(\tau, \cdot) : \Omega \to \mathbb{R}$$

is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

Alternatively, we define a stochastic process as follows:

Definition 1.2. One can equivalently view a stochastic process as a function

$$X : \Omega \times T \to \mathbb{R}, \qquad (\omega, \tau) \mapsto X(\omega, \tau),$$

where:

- For each $\tau \in T$, the mapping $X(\cdot, \tau) : \Omega \to \mathbb{R}$ is measurable with respect to $\mathcal{F}$.

- For each $\omega \in \Omega$, the mapping $X(\omega, \cdot) : T \to \mathbb{R}$ is a trajectory (sample path) of the process.

Key Perspectives

- Random variable view: For each $\tau \in T$, $X(\tau)$ is a random variable.

- Sample path view: For each $\omega \in \Omega$, $X(\omega, \cdot)$ is a deterministic function of time.

- Distributional view: The finite-dimensional distributions $X(\tau_1), \ldots, X(\tau_n)$ for $\tau_1, \ldots, \tau_n \in T$ characterize the process.

Figure 1 illustrates the concept of a stochastic process through the simulation of multiple sample paths of a simple random walk, a canonical example within stochastic process theory. Each trajectory corresponds to a distinct realization of the process, associated with a particular outcome $\omega \in \Omega$. The figure depicts the temporal evolution of the process values across these realizations, thereby exemplifying the intrinsic randomness and variability that are fundamental characteristics of stochastic processes.

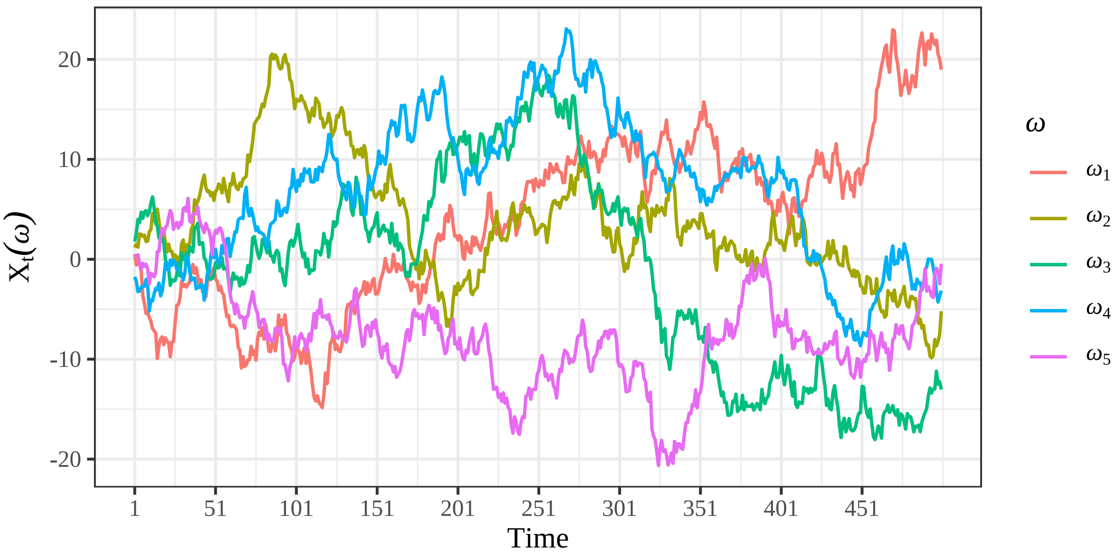

Figure 1: Multiple simple random-walk sample paths (5)

A stochastic process formalizes "random evolution over time": given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, the process assigns a random variable to each time point and, for a fixed outcome ω, a deterministic sample path. Figure 1 illustrates multiple sample paths (realizations) of a simple random walk. Each trajectory corresponds to a distinct realization. The figure depicts the temporal evolution of the process values across these realizations. It exemplifies the intrinsic randomness and variability that are fundamental characteristics of stochastic processes.

2 Stationary Stochastic Processes

A stochastic process is stationary if its statistical properties do not change over time. Stationarity is a fundamental concept in time-series analysis, as many models and inference methods rely on it. There are two main types of stationarity: weak (second-order) stationarity and strong (strict) stationarity.

2.1 Weak Stationarity

Definition 2.1 (Weak (second-order) stationarity)**.** Let $\{X_t : t \in T\}$ be a stochastic process with finite second moments. The process is said to be weakly stationary (or second-order stationary) if:

1. $\mathbb{E}[X_t] = \mu$ for all $t \in T$ (the mean is constant), and 2. $\mathrm{Cov}(X_{t+h}, X_t) = \gamma(h)$ depends only on the lag h (the autocovariance is a function of lag only), for all t, h for which the moments exist.

Equivalently, weak stationarity implies that $\mathrm{Var}(X_t) = \gamma(0)$ is constant over time. Many linear time-series models with finite variance (for example, ARMA models with appropriate coefficients) are weakly stationary. Weak stationarity is the common assumption used in spectral analysis, autocorrelation estimation, and linear forecasting methods.

Weak (or covariance) stationarity refers to a property of a time series whereby, notwithstanding its apparent randomness, its fundamental second-order characteristics remain invariant over time. Specifically, the mean of the series is constant, and the covariance between any two observations depends solely on the temporal distance (lag) separating them rather than on their specific positions in time. Intuitively, if one were to apply a fixed-length moving window across the series and compute summary statistics such as the mean, variance, and autocovariances at various lags (e.g., 1, 2, or 3), these measures would remain stable and exhibit no systematic drift as the window progresses along the time axis. The series does not exhibit systematic increases in level, heightened volatility, or alterations in its correlation structure over calendar time; rather, it persistently replicates an identical variance–covariance configuration. This condition constitutes a less stringent requirement than full (strong) stationarity, as it disregards the complete distributional form and instead concentrates exclusively on the first and second moments. Such a focus is frequently sufficient for linear modeling frameworks and for a wide range of practical time-series methodologies.

The figure below illustrates a simulated weakly stationary AR(1) process, which is a common example of a stationary time series. The plot demonstrates the constancy of the mean (indicated by the red dashed line) and the stable variance over time, visually confirming the properties of weak stationarity.

- Constancy of mean
- Stable variance

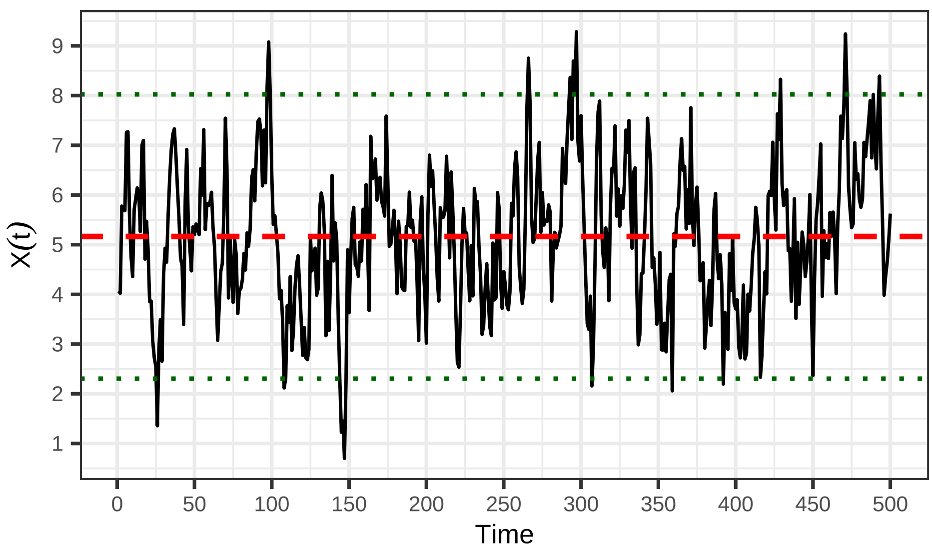

Figure 2: Simulated stationary AR(1) process (mean dashed)

The figure shows a simulated AR(1) series with coefficient $\phi = 0.8$ and unconditional mean approximately $\mu = 5$. The dashed red horizontal line marks the sample mean, and the dotted green lines mark the bands at approximately ± 2 sample standard deviations. Deviations from the mean are persistent but mean-reverting (due to $\phi < 1$), and the overall variance remains roughly constant over time — together these features visually illustrate weak (second-order) stationarity: a constant mean and a covariance structure that does not drift over time.

Remark 2.2. Weak stationarity requires only constant mean and autocovariance, whereas strong stationarity requires full distributional invariance under time shifts. Although strict stationarity is a stronger condition, weak stationarity is often sufficient in practice, especially for linear models and analyses focused on second-order properties.

2.2 Strong Stationarity

A strongly (strictly) stationary process has the same finite-dimensional distributions under any time shift. This is a stronger requirement than weak stationarity, which only constrains the first two moments; strict stationarity demands full distributional invariance.

Definition 2.3 (strong stationarity)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\{X_t : t \in T\}$ be a stochastic process indexed by $T \subseteq \mathbb{R}$. The process $\{X_t\}$ is said to be strongly stationary (or strictly stationary) if, for every integer $n \geq 1$, for every choice of time points $t_1, t_2, \dots, t_n \in T$, and for every shift $h \in \mathbb{R}$ such that $t_i + h \in T$, we have $(X_{t_1}, X_{t_2}, \dots, X_{t_n}) \overset{d}{=} (X_{t_1+h}, X_{t_2+h}, \dots, X_{t_n+h})$, where $\overset{d}{=}$ denotes equality in distribution.

Strong (or strict) stationarity denotes the property whereby the complete probabilistic structure of a stochastic process remains invariant under temporal translation. That is, not only the mean or variance, but the entire joint distribution of any finite collection of time-indexed random variables remains unchanged under any shift of the time index. Intuitively, if one were to extract numerous short segments of the series—such as triplets or longer contiguous subsequences—and randomly permute their temporal order, it would be impossible to determine whether a given segment originated earlier or later in time. This is because all configurations of values, along with their associated probability distributions, remain invariant under temporal translation. This constitutes a highly stringent requirement: it precludes not only the presence of trends and temporal variation in variability, but also any modification in the distributional form or dependence structure of the data. Consequently, the data-generating process is, in a fundamental sense, temporally invariant.

Key Points

• Entire distribution invariance: Not just the mean, variance, or covariance, but the full joint distribution of any finite collection of random variables remains the same under time shifts.

• Stronger than weak stationarity: Weak stationarity only requires invariance of first and second moments, while strong stationarity requires invariance of all finite-dimensional distributions.

• Implication: Every strong stationary process is weakly stationary (if moments exist), but not every weakly stationary process is strongly stationary.

Intuitively, strict stationarity means that time shifts do not change the joint distribution of observations.

Example 2.4. An everyday example is i.i.d. white noise: independence plus identical marginals imply that every finite-dimensional joint distribution is invariant under time shifts, so the process is strictly stationary.

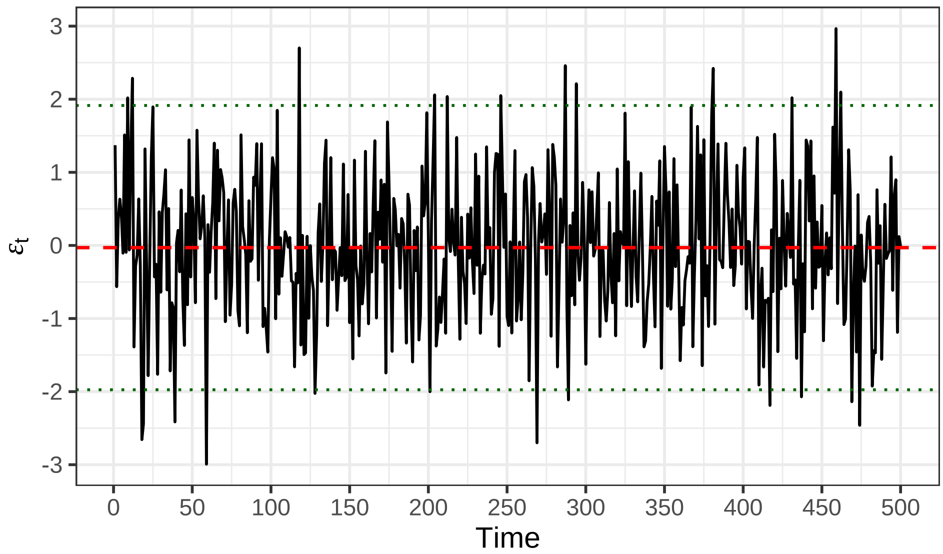

Figure 3: Gaussian white noise (i.i.d. N(0,1))

The figure shows a realization of Gaussian white noise: i.i.d. draws from $\mathcal{N}(0, 1)$. The series has (approximately) zero mean, constant variance, and no serial correlation at nonzero lags. Because the observations are independent with identical marginals, this process is a canonical example of a strictly (strongly) stationary process: every finite-dimensional distribution is invariant under time shifts. It is therefore also weakly stationary (second-order) when second moments exist. This strongly stationary example contrasts with the AR(1) above, where temporal dependence yields persistence while still satisfying weak stationarity whenever $|\phi| < 1$.

Example 2.5. The Bernoulli process (i.i.d. Bernoulli(p)) is likewise strictly stationary.

These examples illustrate that identical marginal distributions and independence across time imply strict stationarity.

A classic example of a process that is weakly stationary but not strongly stationary is a non-Gaussian process with time-invariant mean and covariance (often sharing the same covariance structure as a Gaussian stationary process), since in the non-Gaussian case weak stationarity need not imply strong stationarity. To make it concrete, consider the following:

Example 2.6. Define $X_t = Z_t Y$ with Z_t i.i.d. mean-zero noise and $Y \in \{\pm 1\}$ independent of the Z_t. The process has constant mean and lag-dependent covariance (weak stationarity), and finite-dimensional distributions are invariant under time shifts, so it is strictly stationary, as well.

In short: weak stationarity only cares about mean, variance, and covariance stability, while strong stationarity requires full distributional invariance. This example illustrates how a process can meet the weaker conditions without satisfying the stronger ones.

Table 1 presents a compact summary of the series and models analyzed: each row corresponds to a data series or simulated DGP and columns report the sample size, the sample mean and variance, the estimated AR(1) coefficient (with its standard error), the first few autocovariances/autocorrelations, and the key test statistics with p-values used to assess weak stationarity and unit roots. Together these entries show the degree of persistence and variability for each series, allow quick comparison of estimated dynamics across models (e.g., stationary vs. near-unit-root behavior), and indicate which series reject or fail to reject stationarity at conventional significance levels.

Table 1: Summary comparison of weak (second-order) and strong (strict) stationarity: definitions, requirements, implications, and practical considerations.

Aspect	Weak (second-order) stationarity	Strong (strict) stationarity
Definition	Constant mean and covariance depend only on lag: $\mathbb{E}[X_t] = \mu$ and $\mathrm{Cov}(X_{t+h}, X_t) = \gamma(h)$ for all t, h.	All finite-dimensional distributions are invariant under time shifts: for every k, $t_1, \dots, t_k$ and shift h, $(X_{t_1+h}, \dots, X_{t_k+h}) \overset{d}{=} (X_{t_1}, \dots, X_{t_k})$.
Required moments	Requires existence of first and second moments.	No moment requirement in the definition.
Condition on distributions	Only first two moments must be time-invariant.	Entire joint distributions must be time-invariant.
Implication relation	Does not imply strong stationarity in general.	Implies weak stationarity if first and second moments exist.
Gaussian case	For Gaussian processes, weak $\Leftrightarrow$ strong (mean & covariance determine law).	Same as left column under Gaussianity.
Typical examples	ARMA processes with finite variance.	IID processes; time-homogeneous Markov processes.

Aspect	Weak (second-order) stationarity	Strong (strict) stationarity
Detectability / tests	Tests focus on mean/covariance (ADF, KPSS, sample ACF).	Hard to test directly; compares finite-dimensional distributions.
Practical use	Sufficient for spectral analysis and linear forecasting.	Stronger theoretical guarantees; rarely required in practice.
Vulnerabilities	Can hide nonstationary higher moments or non-Gaussian features.	More robust but harder to verify.

2.3 Examples of Stationary Processes

2.3.1 White Noise

2.3.2 Moving Average (MA) Processes

2.3.3 Autoregressive (AR) Processes

2.3.4 Autoregressive Moving Average (ARMA) Processes

3 Non-stationary Stochastic Processes

3.1 Trend Stationary Processes

3.2 Integrated Processes

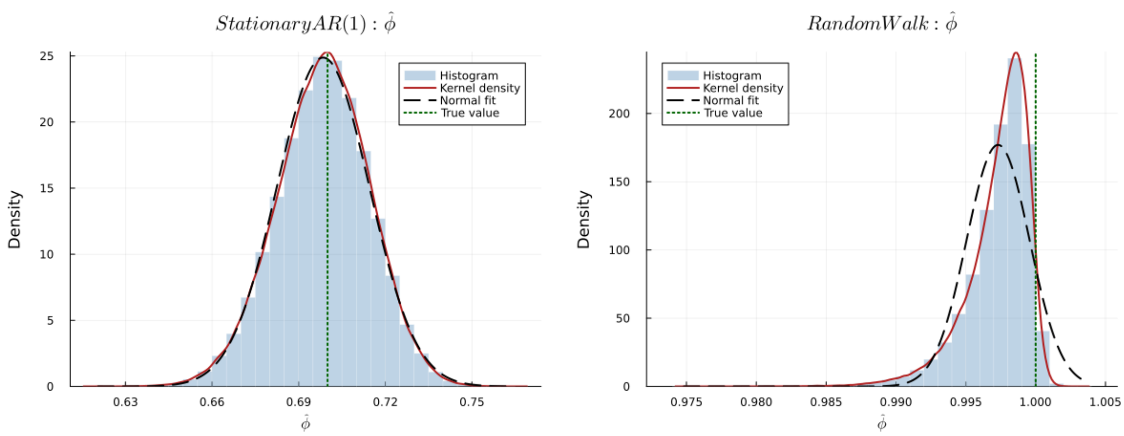

Figure 4: OLS $\widehat{\phi}$ distributions for stationary AR(1) and random walk processes